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Enhanced antibacterial effect of antibiotics in combination with silver nanoparticles against animal pathogens

Monika Smekalova [a], Virginia Aragon [b], Ales Panacek [a,*], Robert Prucek [a], Radek Zboril [a], Libor Kvitek [a]

[a] Regional Centre of Advanced Technologies and Materials, Department of Physical Chemistry, Faculty of Science, Palacky University, 17 Listopadu 12, 771 46 Olomouc, Czech Republic
[b] Centre de Recerca en Sanitat Animal (CReSA), Campus de la Universitat Autònoma de Barcelona, 08193 Bellaterra, Barcelona, Spain

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A B S T R A C T

Antibiotic resistant bacteria are a serious health risk in both human and veterinary medicine. Several studies have shown that silver nanoparticles (AgNPs) exert a high level of antibacterial activity against antibiotic resistant strains in humans. The aim of this study was to evaluate the antibacterial effects of a combined therapy of AgNPs and antibiotics against veterinary bacteria that show resistance to antibiotics. A microdilution checkerboard method was used to determine the minimal inhibitory concentrations of both types of antimicrobials, alone and in combination. The fractional inhibitory concentration index was calculated and used to classify observed collective antibacterial activity as synergistic, additive (only the sum of separate effects of drugs), indifferent (no effect) or antagonistic.

From the 40 performed tests, seven were synergistic, 17 additive and 16 indifferent. None of the tested combinations showed an antagonistic effect. The majority of synergistic effects were observed for combinations of AgNPs given together with gentamicin, but the highest enhancement of antibacterial activity was found with combined therapy together with penicillin G against *Actinobacillus pleuropneumoniae*. *A. pleuropneumoniae* and *Pasteurella multocida* originally resistant to amoxycillin, gentamicin and colistin were sensitive to these antibiotics when combined with AgNPs. The study shows that AgNPs have potential as adjuvants for the treatment of animal bacterial diseases.

Introduction

Extensive use of antimicrobial agents contributes to the development and rapid spread of bacterial resistance, which implies a decrease in antibiotic efficacy in both human and veterinary medicine (Schwarz et al., 2001). Resistance to antimicrobial agents in commensal bacteria (e.g. *Escherichia coli*), zoonotic enteropathogens (e.g. *Salmonella* spp.) and animal pathogens (e.g. *Pasteurella multocida* or *Actinobacillus* spp.) has been reported (Abd-Elghany et al., 2014; Chantziaras et al., 2014; Dayao et al., 2014). One approach to control bacterial infections is combination therapy in which antibiotics are given together with other antimicrobial or non-antimicrobial agents. These adjuvants include other antibiotics, non-antibiotic substances (e.g. cardiovascular drugs), resistance inhibitors, such as β-lactamase inhibitors, and inhibitors of biofilm formation (Kalan and Wright, 2011).

Silver nanoparticles (AgNPs) have bactericidal effects against many species of human bacteria (Kim et al., 2007; Taglietti et al., 2012) and their veterinary counterparts (Soltani et al., 2009), including highly resistant strains, such as methicillin resistant *Staphylococcus aureus* (MRSA) (Panáček et al., 2006; Lara et al., 2009). AgNPs are efficient as antimicrobial agents at low concentrations (mg/L) and are not cytotoxic to eukaryotic cells, including human erythrocytes (Krajewski et al., 2013). Moreover, because of the nonspecific mechanism of AgNP-mediated antibacterial activity, the risk for development of resistance is not as high as for antibiotics.

The mechanism whereby AgNPs interact with bacteria cannot be described in terms of a single and specific mode, as with antibiotics. AgNPs damage the bacterial cell wall, change membrane permeability, and cause a collapse of plasma membrane potential (Sondi and Salopek-Sondi, 2004; Lok et al., 2006). Furthermore, AgNPs interact with DNA, inactivate enzymes, influence metabolic processes, change protein expression and damage the respiratory chain (Lara et al., 2009; Li et al., 2011; Cui et al., 2013). Silver ions are released from the nanoparticle surface and enter the bacterial cell to generate reactive oxygen species (ROS), which destroy biomacromolecules (Choi and Hu, 2008).

AgNPs facilitate the interaction of antibiotics with cells in numerous ways. For example, they may help the penetration of antibiotics into the bacterial cell by changing membrane permeability;

* Corresponding author. Tel.: +420 585 634427.
E-mail address: ales.panacek@upol.cz (A. Panacek).

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alternatively, both AgNP and antibiotic may cooperate in the destruction of the cell wall. In the case of β-lactam antibiotics, AgNPs may inhibit hydrolytic β-lactamases produced by bacteria. Damage and weakness by simultaneous action of antibiotic and AgNP will lead to cell death. Hwang et al. (2012) suggested that this synergism is associated with generation of hydroxyl radicals, the alteration of protective cellular functions and an anti-biofilm potential. Therefore, the combination of antibiotics with AgNPs seems to be a more effective method for enhancing antibiotic efficacy in comparison with other adjuvants currently used in clinical practice. The combination implies reduced antibiotic dose requirements, reduced development of bacterial resistance and increased efficiency of co-administered antibiotics.

Many antibiotics, with differing mechanisms of action, are more effective against human bacteria when combined with AgNPs (Li et al., 2005; Jain et al., 2009; Brown et al., 2012; Ghosh et al., 2012; Hwang et al., 2012). Unfortunately, several of these studies evaluated combined antimicrobial effects using the disc diffusion method employing standard discs with predetermined concentration of antibiotics. In most cases, a zone of inhibition was apparent indicating that the antibiotic was itself effective (Shahverdi et al., 2007; Birla et al., 2009; Fayaz et al., 2010). There would appear to be no reason to combine drugs if a single drug at a specific concentration is effective.

In the present study, we evaluated the synergistic effects of antibiotics administered at doses lower than their minimum inhibitory concentrations (MICs) with AgNPs of two different sizes to test whether it was possible to enhance the antibacterial activity of antibiotics that targeted veterinary-relevant bacteria, including resistant species. Potential synergistic antibacterial effects were quantified by the fractional inhibitory concentration (FIC) index, which was determined using MICs obtained by the microdilution checkerboard method (Lorian, 2005).

Materials and methods

Preparation of silver nanoparticles

AgNPs with diameters of 28 nm (AgNP-28 nm) and 8 nm (AgNP-8 nm) were synthesised through the reduction of complex cation $[Ag(NH_3)_2]^+$ by D-maltose (Sivera et al., 2014) and sodium borohydride, respectively. Detailed description of synthetic procedures and characterisation techniques is included in the Appendix: Supplementary material.

Evaluation of synergistic effect of antibiotics combined with AgNPs

Bacterial strains *Salmonella enterica* LT2, *Staphylococcus aureus* GP0004, *Escherichia coli eae+* GN2514, *Actinobacillus pleuropneumoniae* 17/06L, *Pasteurella multocida* P-813 and *Streptococcus uberis* GP1037 were obtained from the Centre de Recerca en Sanitat Animal, Barcelona, Spain.

The antibiotic sensitivity of bacterial strains was assessed according to Clinical Laboratory Standards Institute (CLSI) document VET01-A3 (CLSI, 2008). MICs of antibiotics, AgNPs and their combinations were determined using the microdilution checkerboard method. *S. enterica*, *S. aureus*, *S. uberis* and *E. coli* were tested in MH (pH 7.3), while *A. pleuropneumoniae* and *P. multocida* were tested in BHI (pH 7.4). BHI was supplemented with 1% IsoVitaleX (BBL) for cultivation of *A. pleuropneumoniae*. The microtitre plates were incubated at 37 °C for 24 h (see Appendix: Supplementary material).

The FIC index was calculated to evaluate the combined antimicrobial effect of antibiotics and AgNPs (Lorian, 2005):

$$\text{FIC} = \frac{\text{MIC of drug A in the combination}}{\text{MIC of drug A alone}} + \frac{\text{MIC of drug B in the combination}}{\text{MIC of drug B alone}}$$

There are differences in the interpretation of the FIC index (Botelho, 2000; Orhan et al., 2005; Kurek et al., 2012). In the present study, the combined antibacterial effects of antibiotics and AgNPs were considered to be synergistic, additive, indifferent or antagonistic when the FIC indices were ≤0.5, >0.5 to ≤1, >1 to ≤2 and >2, respectively (Kurek et al., 2012).

Results

Silver nanoparticles

The results of antibacterial activity of AgNPs are summarised in Table 1. AgNPs exhibited antimicrobial activity against Gram negative organisms. The MICs of AgNPs were 6.3–100 µg/mL depending on the tested strains and size of AgNPs. The growth of Gram positive bacteria was inhibited less by AgNPs than Gram negative bacteria. However, the final MICs of AgNP-28 nm could not be determined because the MIC was greater than the highest silver concentration used in the dilution method (>50 µg/mL).

The antibacterial activity of smaller nanoparticles was stronger than the antibacterial activity of larger AgNPs. Both AgNP-28 nm and AgNP-8 nm showed the most antibacterial effects against the Gram negative bacteria *P. multocida* and *E. coli* (AgNP-8 nm), whereas the Gram positive bacterium *S. uberis* was the least sensitive of the tested strains.

Amoxycillin

The MIC of amoxycillin against the tested bacteria was 0.25 to >32 µg/mL (Table 2). *E. coli* and *A. pleuropneumoniae* had MICs >32 µg/mL (breakpoint MIC ≥32 µg/mL) and were resistant to amoxycillin. When 8 µg/mL amoxycillin was combined with 12.5 µg/mL AgNP-28 nm or AgNP-8 nm, antibiotic resistant *A. pleuropneumoniae* becomes sensitive. The antibacterial activity of amoxycillin combined with AgNP-28 nm was synergistic (FIC index 0.4). An FIC index of 0.6 was obtained when AgNP-8 nm was combined with

Table 1
Minimum inhibitory concentrations (µg/mL) of silver nanoparticles with diameters of 28 nm (AgNP-28 nm) or 8 nm (AgNP-8 nm).

Bacteria	AgNP-28 nm	AgNP-8 nm
Salmonella enterica	25	12.5
Staphylococcus aureus	>50	25
Escherichia coli	25	6.3
Actinobacillus pleuropneumoniae	50	25
Streptococcus uberis	>50	100
Pasteurella multocida	6.3	6.3

Table 2
Minimum inhibitory concentrations (MICs; µg/mL) of amoxycillin alone or with silver nanoparticles with diameters of 28 nm (AgNP-28 nm) or 8 nm (AgNP-8 nm).

Bacteria	Amoxycillin	Drug combination MIC		FIC	Effect	Drug combination MIC		FIC	Effect
		AgNP-28 nm	Amoxycillin			AgNP-8 nm	Amoxycillin		
Salmonella enterica	1	25	1	2.0	I	12.5	1	2.0	I
Staphylococcus aureus	2	25	1	0.8	A	12.5	0.5	0.8	A
Escherichia coli (R)	>32	25	>32	2.0	I	6.3	>32	2.0	I
Actinobacillus pleuropneumoniae (R)	>32	12.5	8	0.4	S	12.5	8	0.6	A
Pasteurella multocida	0.25	6.3	0.25	2.0	I	6.3	0.25	2.0	I
Streptococcus uberis	0.25	>50	0.25	2.0	I	50	0.125	1.0	A

FIC, fractional inhibitory concentration index; S, synergy; A, additivity; I, indifference; R, resistant strain.

Table 3
Minimum inhibitory concentrations (MICs; µg/mL) of penicillin G alone or with silver nanoparticles with diameters of 28 nm (AgNP-28 nm) or 8 nm (AgNP-8 nm).

Bacteria	Penicillin G	Drug combination MIC		FIC	Effect	Drug combination MIC		FIC	Effect
		AgNP-28 nm	Penicillin G			AgNP-8 nm	Penicillin G		
Salmonella enterica	2	25	1	1.5	I	12.5	1	1.5	I
Staphylococcus aureus (R)	1	12.5	0.5	0.6	A	12.5	0.125	0.6	A
Escherichia coli (R)	>32	25	>32	2.0	I	6.3	>32	2.0	I
Actinobacillus pleuropneumoniae (R)	>32	25	8	0.6	A	6.3	2	0.3	S
Pasteurella multocida	0.03	6.3	0.03	2.0	I	6.3	0.03	2.0	I

FIC, fractional inhibitory concentration index; S, synergy; A, additivity; I, indifference; R, resistant strain.

Table 4
Minimum inhibitory concentrations (MICs; µg/mL) of gentamicin alone or with silver nanoparticles with diameters of 28 nm (AgNP-28 nm) or 8 nm (AgNP-8 nm).

Bacteria	Gentamicin	Drug combination MIC		FIC	Effect	Drug combination MIC		FIC	Effect
		AgNP-28 nm	Gentamicin			AgNP-8 nm	Gentamicin		
Salmonella enterica	1	12.5	0.25	0.8	A	12.5	0.25	1.3	I
Staphylococcus aureus	2	25	0.25	0.4	S	6.3	0.5	0.5	S
Escherichia coli	>32	6.3	8	0.4	S	3.1	16	0.8	A
Actinobacillus pleuropneumoniae (R)	8	25	0.25	0.5	S	25	8	2.0	I
Pasteurella multocida (R)	8	3.1	2	0.8	A	3.1	2	0.8	A

FIC, fractional inhibitory concentration index; S, synergy; A, additivity; I, indifference; R, resistant strain.

Table 5
Minimum inhibitory concentrations (MICs; µg/mL) of colistin alone or with silver nanoparticles with diameters of 28 nm (AgNP-28 nm) or 8 nm (AgNP-8 nm).

Bacteria	Colistin	Drug combination MIC		FIC	Effect	Drug combination MIC		FIC	Effect
		AgNP-28 nm	Colistin			AgNP-8 nm	Colistin		
Salmonella enterica (R)	4	12.5	2	1.0	A	6.3	2	1.0	A
Escherichia coli (R)	8	12.5	2	0.8	A	1.6	2	0.5	S
Actinobacillus pleuropneumoniae (R)	8	25	2	0.8	A	25	8	2.0	I
Pasteurella multocida (R)	8	3.1	1	0.6	A	3.1	1	0.6	A

FIC, fractional inhibitory concentration index; S, synergy; A, additivity; I, indifference; R, resistant strain.

amoxycillin. The additive effects of amoxycillin in combination with both sizes of AgNPs were observed against Gram positive *S. aureus*, with an FIC index of 0.8. An additive effect with an FIC index of 1.0 was also demonstrated for amoxycillin combined with AgNP-8 nm against *S. uberis*.

Penicillin G

Penicillin G exhibited a similar pattern of results to amoxycillin (Table 3). The MIC values for penicillin G were 0.03 to >32 µg/mL. *S. aureus* (MIC 1 µg/mL, breakpoint MIC ≥0.25 µg/mL), *E. coli* (MIC >32 µg/mL, breakpoint MIC ≥4 µg/mL) and *A. pleuropneumoniae* (MIC >32 µg/mL, breakpoint MIC ≥4 µg/mL) showed resistance to penicillin G. The antibacterial effects of penicillin G in combination with AgNP-8 nm or AgNP-28 nm were synergistic and additive, respectively, against *A. pleuropneumoniae*. This synergistic effect was the strongest of all combinations tested, with an FIC index of 0.3 for the combination of 2 µg/mL penicillin G plus 6.3 µg/mL AgNP-8 nm. Additive effects were observed for *S. aureus* with penicillin G plus AgNPs of either size.

Gentamicin

The most effective antibiotic in combination with AgNPs was gentamicin (Table 4). When gentamicin was combined with AgNP-8 nm, enhanced antibacterial effects were additive against *S. enterica* and *P. multocida*, and synergistic against *S. aureus*, *E. coli* and *A. pleuropneumoniae*. With AgNP-8 nm, the effects were additive against *E. coli* and *P. multocida*, and synergistic against *S. aureus*. Combining 3.1 µg/mL of either AgNP-28 nm or AgNP-8 nm with gentamicin enhanced sensitivity of resistant *P. multocida* (MIC 8 µg/mL, breakpoint MIC ≥8 µg/mL), sufficient to make these bacteria gentamicin-sensitive. Resistant *A. pleuropneumoniae* (MIC 8 µg/mL, breakpoint MIC ≥8 µg/mL) became sensitive to gentamicin when this antibiotic was combined with AgNP-28 nm.

Colistin

The MICs of colistin reached 4 or 8 µg/mL (Table 5), indicating that all bacteria tested were resistant to this antibiotic (breakpoint MIC ≥2 µg/mL). When combined with AgNP-28 nm, the effect was additive in all cases. Additive effects were also observed for the combination of AgNP-8 nm and colistin against *S. enterica* and *P. multocida*. *P. multocida* also became sensitive to colistin when it was combined with 6.3 µg/mL AgNP-28 nm or 3.1 µg/mL AgNP-8 nm. The combination of colistin with 1.6 µg/mL AgNP-8 nm produced a synergistic effect against *E. coli.*

Discussion

AgNPs are known for their exceptional antibacterial activity against human bacteria (Panáček et al., 2006) and demonstrate synergistic action when given in combination with various adjuvants (Potara et al., 2011). Further investigation into their activity is required to combat fast-spreading antibiotic resistances in bacteria causing important diseases in livestock, such as mastitis (*S. uberis*) or respiratory diseases (*A. pleuropneumoniae*) (Matter et al., 2007). In the present study, AgNPs in combination with four different antibiotics were investigated for efficacy in several bacteria of veterinary origin to determine whether the antibacterial activity of a given antibiotic is enhanced.

The antibacterial activity of AgNPs strongly depends on their size (Martínez-Castañón et al., 2008). In the present study, AgNPs of two different sizes were synthesised to determine whether appropriate enhancement of antibacterial activity of antibiotics combined with AgNPs depends on nanoparticle size. The average sizes of AgNPs preliminary determined by DLS (28 and 8 nm) were larger than those calculated from TEM micrographs (20.6 and 4.7 nm; Appendix: Supplementary Figs. S1A and S2A); the DLS method overestimates larger size fractions, since the intensity of scattered light depends on the sixth power of the particle size. Using UV/Vis spectrophotometry, AgNP-28 nm and AgNP-8 nm showed plasmon bands at 409 nm (Appendix: Supplementary Fig. S1B) and 392 nm (Appendix: Supplementary Fig. S2B), respectively, which are characteristic for nanoscale silver (Kvítek et al., 2005). Smaller nanoparticles present absorption maxima at shorter wavelengths, whereas red shifts in maxima occur when the particle size increases (Panáček et al., 2014).

When non-stabilised AgNP-28 nm particles were mixed with broth prior to bacterial cultivation, slight aggregation of AgNPs occurred, which was evident in the emergence of secondary absorption peaks at longer wavelengths. Secondary absorption peaks located at the wavelengths of 550–650 nm occur when AgNPs undergo aggregation (Kvítek et al., 2008). To prevent this particle aggregation, gelatine was used as a stabilising agent for AgNP-28 nm. Because BHI broth consists of higher concentrations of electrolytes and proteins than MH broth, it was necessary to use a higher gelatine concentration to sufficiently stabilise AgNP-28 nm. In contrast, AgNP-8 nm particles are stabilised by PAA directly in the preparation process and were sufficiently stable in the cultivating media.

The smaller nanoparticles showed greater activity, corresponding to previous observations that antibacterial activity is size-dependent (Martínez-Castañón et al., 2008). Small nanoparticles can pass through pores in the cellular membrane, release greater numbers of silver ions and generate more ROS (Carlson et al., 2008) and the larger surface area to volume ratio makes smaller AgNPs more reactive (Rao et al., 2002). It was also observed that the growth of Gram positive bacteria was less inhibited by AgNPs than Gram negative bacteria, in agreement with previous reports (Kim et al., 2007). These results perhaps can be attributed to the thinner peptidoglycan layer of Gram negative bacteria, in comparison with the rigid peptidoglycan cell wall of Gram positive bacteria.

Amoxycillin can be used as an example of the potential for AgNPs. Amoxycillin is a β-lactam antibiotic, which acts by inhibiting the synthesis of bacterial cell walls (Tipper, 1985) and is used for the treatment of a wide range of bacterial infections in many animal species (Wilson et al., 1999). As an example, amoxycillin has been used against *A. pleuropneumoniae*, which causes important respiratory diseases that result in economic losses in pigs (Bossé et al., 2002). Unfortunately, bacterial strains resistant to amoxycillin have emerged. In the present study we show that the combination of amoxycillin with the two types of AgNPs enhanced its activity against this *A. pleuropneumoniae* such that it became sensitive to amoxycillin. The effects of amoxycillin combined with AgNP-28 nm and AgNP-8 nm were synergistic and additive, respectively, against *A. pleuropneumoniae*. Additive effects were also observed against *S. aureus*, similar to the effects noted by Shahverdi et al. (2007).

Li et al. (2005) proposed possible mechanisms for the synergistic action of amoxycillin and AgNPs. Antibiotics most likely bind to AgNPs, forming a structure in which amoxycillin molecules surround the AgNP core. This favours the concentration of antimicrobial groups at particular points on the cell surface, increasing the severity of damage to the bacterium. In addition, AgNPs facilitate recruitment of amoxycillin to the bacterial cell surface (Fayaz et al., 2010). Brown et al. (2012) showed that the biocidal activity of ampicillin increased when the drug was combined with AgNPs.

Similar results to those observed in the combination assay for amoxycillin were observed with penicillin G, which also belongs to β-lactam group. The strongest synergistic effect in this study was observed for penicillin G combined with AgNP-8 nm directed against *A. pleuropneumoniae*. The additive activity of penicillin G in combination with AgNP-28 nm and AgNP-8 nm against *S. aureus* was in agreement with findings by Shahverdi et al. (2007).

The most enhanced antibacterial activity through combination with AgNPs was achieved with gentamicin. As with other aminoglycosides, gentamicin acts by irreversibly binding to the 30S subunit of the bacterial ribosome so interfering with protein synthesis. In total, four additive and four synergistic effects were observed by combining this antibiotic with AgNPs. Birla et al. (2009) demonstrated an increased activity of gentamicin in the presence of AgNPs against *E. coli* and *S. aureus*. In contrast, the efficiency of a combined AgNP-gentamicin treatment against these bacteria was not demonstrated in other studies (Shahverdi et al., 2007; Ghosh et al., 2012). An important observation from this current study is the sensitivity of *A. pleuropneumoniae* and *P. multocida* to gentamicin in the presence of AgNPs, whereas these strains were resistant to gentamicin alone. *P. multocida* is a Gram negative bacterium that infects wild and domesticated animals and causes important diseases such as fowl cholera (Harper et al., 2006).

The same phenomenon of antibiotic sensitivity of *P. multocida* was observed for combinations of AgNPs with colistin, a mixture of cyclic polypeptides effective against most Gram negative bacteria, whose antibacterial activity is directed against the bacterial cell membrane (Falagas and Kasiakou, 2005).

Most studies dealing with the synergistic effect of antibiotics combined with AgNPs were performed using the disc diffusion method, but this method has no parameter to classify results as synergistic (Shahverdi et al., 2007; Birla et al., 2009; Fayaz et al., 2010; Ghosh et al., 2012). There are also questions as to the extent to which AgNPs are able to diffuse through solid agar and, for the purposes of synergy testing, drugs at concentrations lower than the MICs should be used. Unfortunately, previous studies used each drug in such an amount that evidently inhibits bacterial growth (Shahverdi et al., 2007; Birla et al., 2009; Fayaz et al., 2010).

AgNP-containing materials have been applied for use in human medicine for wound dressing, impregnation of catheters, as implants and in surgical mesh (Nirmala et al., 2010; Antonelli et al., 2012; Wu et al., 2014; Nisticò et al., 2015). Most silver-based products for animals are designed for surface disinfection, but some preparations (e.g. TraumaPet Ag, Auxivet) are used as drops, ointments, sprays or gels to suppress bacterial infections. AgNPs are also used as prebiotics in animal feeding (Fondevila et al., 2009). Effects on organisms depend on specific AgNP characteristics, such as size, morphology or surface modification, and different AgNPs show different level of toxicity. Although potential adverse health implications associated with AgNPs have been reported (Korani et al., 2011; Genter et al., 2012; Patlolla et al., 2015), several cytotoxic studies demonstrated that AgNPs have a wide therapeutic window (Martínez-Gutierrez et al., 2012; Mohanty et al., 2012; Boonkaew et al., 2014; Ivask et al., 2014). Moreover, the combination of AgNPs with antibiotics lowers the concentrations of both antibiotics and silver required to kill bacteria.

Ivask et al. (2014) demonstrated no cytotoxicity to mammalian cells at 26.7 mg/L, while the average EC_{50} (half-maximum effective concentration) for *E. coli* was 1.56 mg/L. AgNPs at 2 μM killed >98% of bacteria, whereas this concentration was harmless for macrophages (Mohanty et al., 2012). AgNP-containing hydrogels designed for infected burn care also showed antimicrobial activity without cytotoxicity (Boonkaew et al., 2014). Martínez-Gutierrez et al. (2012) tested the antibacterial activity of AgNPs against a range of Gram positive and Gram negative bacteria, along with toxicity to THP-1 monocytes; AgNPs had MICs of 0.4–1.7 µg/mL, while a significant increase in cytotoxicity was observed >5 µg/mL.

In mice, the lowest observed adverse effect was at 0.5 mg/kg bodyweight/day following 28 days of repeated oral administration of AgNPs (Park et al., 2010). Conversely, nanoparticles caused only slight liver damage when rats were exposed to >300 mg/kg AgNPs (Kim et al., 2008).

Conclusions

Antibiotics in combination with AgNPs showed synergistic, additive or indifferent effects independent of nanoparticle size, with no antagonism observed. Although most of the effects were additive or indifferent, the combination of AgNPs with antibiotics reduced the MIC of a given antibiotic in most cases. In the case of *P. multocida* and *A. pleuropneumoniae*, AgNPs induced antibiotic sensitivity of the selected strains to amoxycillin, gentamicin and colistin. AgNPs most frequently exhibited the effect by increasing the antibacterial activity of gentamicin, although the strongest synergistic effect was observed for penicillin G combined with AgNP-8 nm against *A. pleuropneumoniae*. The combination of antimicrobial agents with AgNPs could be a promising way to decrease the amount of antibiotics currently used in animal husbandry.

Conflict of interest statement

None of the authors of this paper have a financial or personal relationship with other people or organisations that could inappropriately influence or bias the content of the paper.

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Appendix: Supplementary material

Supplementary data to this article can be found online at doi:10.1016/j.tvjl.2015.10.032.

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